Question ID: 3a9d60b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$2|4-x|+3|4-x|=25$$

What is the positive solution to the given equation?

Correct Answer: 9

Rationale

The correct answer is 9. The given equation can be rewritten as $5|4-x|=25$. Dividing each side of this equation by 5 yields $|4-x|=5$. By the definition of absolute value, if $|4-x|=5$, then $4-x=5$ or $4-x=-5$. Subtracting 4 from each side of the equation $4-x=5$ yields $-x=1$. Dividing each side of this equation by -1 yields $x=-1$. Similarly, subtracting 4 from each side of the equation $4-x=-5$ yields $-x=-9$. Dividing each side of this equation by -1 yields $x=9$. Therefore, since the two solutions to the given equation are -1 and 9, the positive solution to the given equation is 9.